

C

Theory

1. Data Types in C

- **Basic types:** int, float, double, char, bool.
- **Size:**
 - int → usually 4 bytes.
 - float → 4 bytes.
 - double → 8 bytes.
 - char → 1 byte.
- **Note:** Assigning 10.00 to int truncates decimals.

2. Operators in C

- **Unary** → needs one operand (++ , --).
- **Binary** → needs two operands. Categories:
 - Arithmetic (+, -, *, /, %).
 - Relational (>, <, >=, <=, ==, !=).
 - Logical (&&, ||, !).
 - Assignment (+=, -=, *=, /=, %=).
 - Bitwise (&, |, ^, >>, <<).
- **Ternary** → condition ? expr1 : expr2.

3. Control Structures

- **Conditional:** if, else if, else, switch.
- **Loops:** for, while, do-while.
- **Nested if:** allows multiple conditions.

4. Input/Output

- printf() → output.
- scanf() → input.
- Format specifiers: %d (int), %f (float), %c (char), %lf (double).

5. History & Use Cases

- **C language** created in 1972 by Dennis Ritchie.
- **Use cases:** Operating systems, embedded systems, compilers.
- **Strengths:** Fast, close to hardware, portable.

Coding Round Questions

- Data types → Explain different data types and their sizes.
- Operators → Write a program to demonstrate arithmetic, relational, logical, and bitwise operators.
- Control structures → Write a program using nested if to check voting eligibility.
- Input/Output → Write a program to take user input and print it.
- Unary operators → Show how ++ and -- work.
- Assignment operators → Write a program using +=, -=, *=, /=.

⚡ Machine Coding Round Challenge

Problem: Student Result System

Requirements:

- Input student details: name, roll number, marks in 3 subjects.
- Calculate total and average.
- Use conditional statements to assign grade:
 - ≥ 75 → Distinction.
 - ≥ 50 → Pass.
 - < 50 → Fail.
- Display results in a formatted output.

```
#include <stdio.h>
#include <stdbool.h>

int main(){
```

```
int data ; // default value is 0

// printf("Enter integer number: ");
// scanf("%d", &data);

// printf("%d\n", data);

// float marks ;

// printf("Enter float value: ");
// scanf("%f", &marks);

// printf("%.2f\n", marks);

// operators => unary, binary, ternary

// unary => ++, --
// binary => arithmetic, relational, logical, assignment, bitwise

// relational => > , < , >=, <=, !=, ==

int a = 10;
int b = 10;

// printf("%d \n", a > b);

// printf("%d \n", a < b);

// printf("%d \n", a >= b);

// printf("%d \n", a <= b);

// printf("%d \n ", a != b);

// printf("%d\n", a == b );

// logical operators  => && || !

// int result = a>b && a<b;

// result = a==b && a<=b;

// result = a==b || a<b;
```

```

// result = !a<b;

// printf("%d\n", result);


// & => true true => true

// || => true false => true
//      false true => true
//      true true => true

//! => true -> false
// false -> true


// assignment operator => += , -=, /=, *=, %=

// int c = 50;

// c += 10;
// c -= 10;
// c *= 10;
// c /= 10;
// c %= 10;

// printf("%d\n", c);

// bitwise operator => &, |, ^, >> ,<<

//      int i=4;
//      int j=7;

//      int resultData = i & j;
//      printf("%d\n",resultData );

//      resultData = i | j;
//      printf("%d\n",resultData );

//      resultData = i ^ j;
//      printf("%d\n",resultData );

```

```

//      resultData = i>>2;
//      printf("%d\n",resultData );

//      resultData = i <<1;
//      printf("%d\n",resultData );


//control structure

// conditional statement, loops

// conditional statement
// if, if-else, else-if, nested if , switch


int age = 30;
// if(age > 18 ){
//     printf("You are eligible to votte");
// }else{
//     printf("You are not eligible to votte");
// }

bool isEligible = false;

if(age >= 18){

    if(isEligible){

        printf("You are eligible to vote");
    }else{
        printf("you don't have voter id");
    }

}

}else{
    printf("You are age is less then 18");
}

return 0;
}

```